

Microbiota-based biomarkers and therapeutics for cancer management

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New light is being shed on the interactions between the gut microbiome, cancer cell signalling and the host immune response. With this knowledge, microbiota-based approaches for improving cancer prevention, prognostication and therapy have started to materialize and will contribute to reducing the global cancer burden.

The continued expansion of multi-omic datasets and their integrative analysis with clinicopathological characteristics has ushered in the rapid transition of cancer microbiome research from the straightforward identification of cancer-specific microorganisms to more mechanistic and clinically oriented studies. In 2023, we witnessed major breakthroughs in cancer microbiome research that could readily – or have a high potential to – translate into clinical benefits in different aspects of cancer management, including screening, prevention, molecular typing and therapy.

Typing of colorectal cancer (CRC) is an evolving field in which different molecular features associated with clinical outcomes have been identified¹. Despite these advances, prognostication in clinical settings still heavily relies on the conventional tumour–node–metastasis (TNM) staging system and the detection of deficient DNA mismatch repair (dMMR) or microsatellite instability (MSI)². Integrative analysis of molecular prognostic features from various omic platforms is anticipated to advance the clinical development of biomarkers for stratification of patients with CRC. To accomplish this goal, Roelands et al. comprehensively characterized the genomic, transcriptomic, immunological and microbial features of a cohort of 348 patients with CRC with long-term follow up data to devise a composite score for prognostication³. The authors first demonstrated that a 20-immune gene signature known as the Immunologic Constant of Rejection (ICR) outperformed existing molecular classifications (including dMMR, MSI and the consensus molecular subtypes (CMS))⁴ for predicting patients' survival probability. Cross-omic analysis revealed that ICR was positively correlated with tumour-infiltrating, clonally expanded T cells and negatively correlated with *MAP3K1*, *BRCA1*, *BRCA2* and *FANCA* mutations. In a subset of 246 patients with paired tumour and normal colon samples, 16S rRNA gene and whole-genome sequencing data further identified a microbial signature consisting of 41 bacterial taxa that could predict overall survival, with the intratumoural abundance of *Ruminococcus bromii* having the strongest contribution to the classifier. The prognostic performance of the microbial signature was independent of age, CMS, TNM stage and MSI status. Combining ICR with the microbial signature into a composite score termed mICRoScore maximized the prognostic performance, in which a mICRoScore-high

group corresponded to a subgroup of patients with CRC with a 97% 5-year overall survival³. This study not only places the intratumoural microbiome centre stage for the development of CRC prognostic markers, but also raises the possibility of using *R. bromii* to enhance antigen processing and presentation during immunotherapy. Nevertheless, the latter remains to be tested in animal models.

Carnobacterium maltaromaticum is one of the most depleted bacteria in the stool samples of patients with CRC, as revealed by our multi-cohort study⁵. Bacterial abundance analysis using The Cancer Genome Atlas dataset further showed that the intratumoural depletion of *C. maltaromaticum* was associated with worse overall survival in patients with CRC, especially among female patients⁶. Our laboratory was thus motivated to examine its role in CRC development. In two murine models of intestinal tumorigenesis, *C. maltaromaticum* reduced tumour number and tumour load in a female-specific manner. *C. maltaromaticum* abundance was higher in females than in males in both humans and mice; in the latter, its colonization and the resulting anti-CRC effect could be abrogated by ovariectomy in females or rescued by ovariectomy plus 17 β -oestradiol administration in males. An oestrogen-responsive host receptor, SLC3A2, was subsequently identified to mediate the female-biased colonization of *C. maltaromaticum* via binding to the bacterial D-alanyl-D-alanine carboxypeptidase.

High dietary vitamin D consumption is estimated to reduce CRC risk by ~12%⁷. Synthesis in the skin from 7-dehydrocholesterol (7-DHC) through sun exposure and absorption via consumption of certain foods (for example, fatty fish, fortified dairy and eggs) are two known major sources of vitamin D. In *C. maltaromaticum*-gavaged mice, vitamin D metabolites were enriched in the gut, leading to mucosal activation of vitamin D receptor signalling. Interestingly, *C. maltaromaticum* was found not to be able to produce vitamin D by itself. Instead, it secreted 7-DHC to enhance the growth of *Faecalibacterium prausnitzii* and *Lachnospiraceae bacterium*, both of which converted 7-DHC into 25-OH and/or 1 α ,25-(OH)₂ vitamin D₃ to mediate vitamin D receptor activation for CRC suppression⁶. Our study highlights the complexity of metabolic crosstalk within the intestinal microbial community and the role of

Key advances

- A *Ruminococcus bromii*-driven microbiome signature that correlates with immune traits predicts overall survival of patients with colorectal cancer³.
- *Carnobacterium maltaromaticum* is a potential prophylactic for colorectal cancer via enhancing intestinal vitamin D biogenesis⁶.
- Downregulation of PDL2 by the gut microbiome, and specifically *Coprobacillus cateniformis*, is required for the efficacy of anti-PD1 or anti-PDL1 immunotherapy¹⁰.

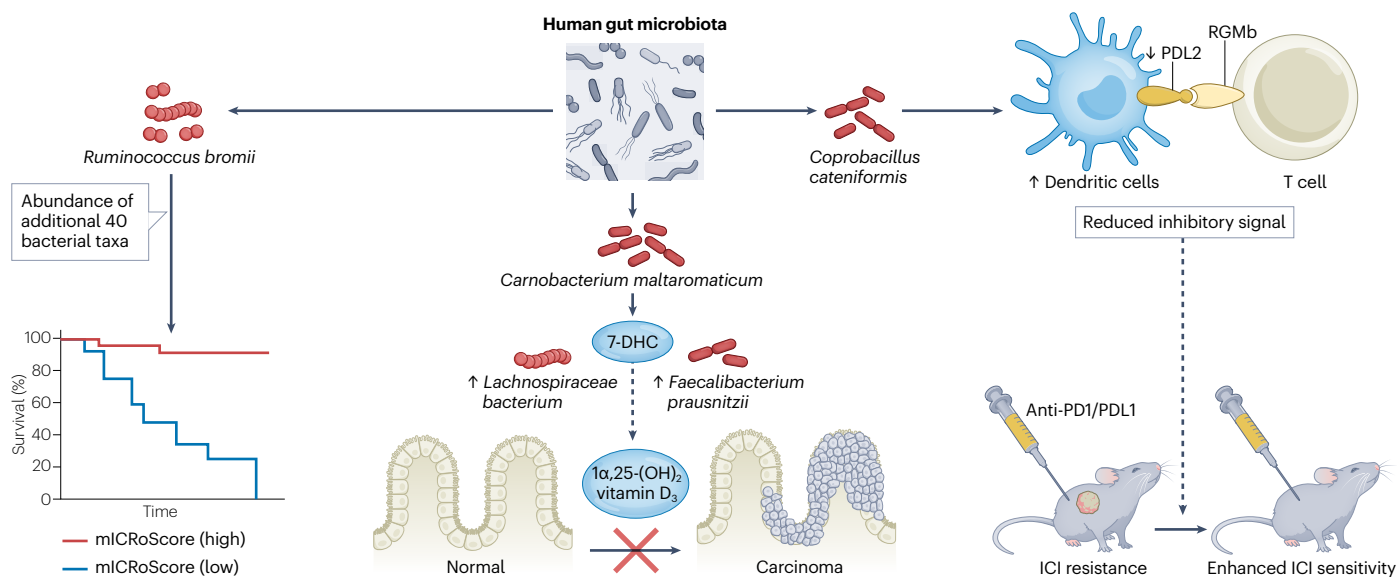


Fig. 1 | Specific microorganisms from the human gut microbiota as emerging prognostic markers and therapeutics. Combining a *Ruminococcus bromii*-driven microbiome signature and the expression of 20 immune genes into a composite score (mICRoScore) stratifies patients with colorectal cancer (CRC) with disparate survival probability. *Carnobacterium maltaromaticum*

secretes 7-dehydrocholesterol (7-DHC) for conversion by *Faecalibacterium prausnitzii* and *Lachnospiraceae bacterium* into bioactive vitamin D metabolites for CRC suppression. *Coprobacillus cateniformis* downregulates the expression of the co-inhibitory molecule PDL2 in dendritic cells to overcome anti-PD1/PDL1 resistance. ICI, immune checkpoint inhibitor; RGMb, repulsive guidance molecule b.

the gut microbiota as an alternative source of vitamin D. To this end, *C. maltaromaticum* holds promise to be developed as a prophylactic for CRC prevention in females.

The ability of faecal microbiota transplantation from immune checkpoint inhibitor (ICI) responders to non-responders to restore ICI responsiveness has sparked intense interest for the identification of specific gut microorganisms or their functional components that mediate the adjuvant effect^{8,9}. Park et al. demonstrated the importance of the gut microbiota in ICI responsiveness by showing that tumours implanted subcutaneously into germ-free or antibiotic-treated mice no longer responded to anti-programmed death 1 ligand 1 (PDL1) or anti-programmed death 1 (PD1) treatment¹⁰. Reconstitution with healthy human microbiota restored the anti-PD1 and/or anti-PDL1 responsiveness in these mice, paralleled by the early expansion of MHC-II⁺CD11b⁺ and MHC-II⁺CD11c⁺ dendritic cells in the tumour-draining lymph nodes and downregulation of the immune checkpoint molecule PDL2 in these cell populations. The functional importance of PDL2 downregulation in anti-PD1 and anti-PDL1 responsiveness was corroborated by the potentiating effect of anti-PDL2 on the efficacy of anti-PD1 and anti-PDL1, irrespective of the status of the gut microbiota (germ-free, antibiotic-treated, specific-pathogen-free or transplanted with faecal samples from anti-PD1 non-responders). Through strenuous antibiotic selection, sequencing and functional experiments, *Coprobacillus cateniformis* was pinpointed as the bacterium responsible for the PDL2-downregulating effect of healthy human microbiota and the consequential enhancement of anti-PD1 and/or anti-PDL1 efficacy. Repulsive guidance molecule b (RGMb) on T cells, which was also downregulated upon healthy human microbiota transplantation, was identified as the PDL2-interacting partner to mediate the inhibitory signal¹⁰. This work raises the possibility that *C. cateniformis* supplementation might enhance the efficacy of anti-PD1 or anti-PDL1 in humans via targeting the PDL2–RGMb axis.

Although the development of microbiota-based biomarkers and therapeutics as highlighted above (Fig. 1) seems propitious, given the well-known inter- and intra-population heterogeneity of gut microbiota, testing in international multi-centre trials is essential for promoting clinical utilization. The integration of gut microbiota profiles with additional omic platforms (such as single-cell and spatial transcriptomics,

high-dimensional flow and mass cytometry) will hopefully bring more discoveries to discern the intricate role of the gut microbiota in cancer.

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Competing interests

The authors declare no competing interests.